RFC-HMS-2026-002

Workflow Engine, Cross-Module Automation & Hospital Configuration Architecture

7-Layer Configuration Engine | 120+ Configurable Workflow Templates

250+ Cross-Module Automation Links | Visual Workflow Builder

Dynamic Form Engine | Rules Engine | Hospital Onboarding Framework

Companion to: RFC-HMS-2026-001 v2.0 (Core + ABAC) & v2.1 (Extended Modules)

Version 1.0 | March 2026 | Alagappa Group of Institutions | CONFIDENTIAL

Table of Contents

1. Architecture Overview: Why Configuration Over Code

Problem: Every hospital is different. Different buildings, different department structures, different workflows, different approval hierarchies, different billing rules, different equipment, different integrations. If we hard-code any of these, every new hospital deployment requires code changes = developer dependency = unscalable.

Solution: A 7-layer configuration engine where the application code is FIXED (same binary for all hospitals), but behavior is driven entirely by configuration data stored per-tenant. Hospital administrators configure the system through visual tools — no coding required.

This RFC defines the complete configuration architecture, every configurable workflow template across all 67 modules, and every automated cross-module link that makes the system feel like a living, intelligent organism rather than disconnected screens.

Layer	Name	What It Configures	Configuration Tool
Layer 1	Hospital Topology	Buildings, floors, wings, rooms, beds, equipment locations, gas outlets, power zones	Visual floor plan editor + tree builder
Layer 2	Department & Services	Department hierarchy, sub-departments, units, service catalog, procedure list, investigation menu	Tree builder + service catalog manager
Layer 3	Workflow Engine	Every clinical & admin process: step sequence, branching, approvals, SLAs, escalations, auto-actions	Visual drag-and-drop workflow builder
Layer 4	Form Builder	Data capture screens: fields, validations, conditional logic, calculations, templates per specialty	Drag-and-drop form designer
Layer 5	Rules Engine	Business rules: clinical thresholds, financial limits, approval chains, alerts, compliance rules	Rule builder with IF-THEN-ELSE logic
Layer 6	Billing Configurator	Tariff master, rate schemes, packages, tax rules, payment modes, insurance mapping, government schemes	Rate card manager + package builder
Layer 7	Integration Hub	Lab analyzers, accounting systems, government portals, communication channels, medical devices, PACS	Integration panel with connector marketplace

Key Principle: Layers 1-2 are configured once during onboarding (rarely changed). Layers 3-5 are living configurations — hospital can modify workflows and rules as their processes evolve. Layers 6-7 are configured during onboarding and updated when tariffs change or new equipment arrives.

2. Layer 1: Hospital Topology Configurator

2.1 Physical Layout Data Model

The system models every hospital as a hierarchical tree of physical spaces. This is the foundation — bed management, patient location, equipment tracking, ABAC context-awareness, nurse call routing, and emergency response all depend on this.

Level	Entity	Attributes	Example
L0	Campus	Campus name, address, GPS coordinates, total area, emergency assembly points, helipad (Y/N)	Alagappa Medical College Campus, Karaikudi
L1	Building	Building name/code, floors count, year built, structural type, fire escape count, elevator count, DG backup zone	Main Hospital Block (MHB), Teaching Block (TB), Hostel Block (HB)
L2	Floor	Floor number/name, departments on floor, total rooms, fire zone, evacuation route, nurse station count	MHB Floor 3 = ICU + CCU + HDU
L3	Wing	Wing name/code, department, gender restriction, isolation wing (Y/N), barrier nursing (Y/N)	MHB-F3-Wing A = Medical ICU (Mixed), Wing B = Surgical ICU (Mixed)
L4	Zone	Zone type (patient care / nurse station / doctor room / utility / storage), MGPS availability (O2/suction/air), power backup tier	Zone A1 = 10-bed patient area with O2+suction+air at every bed
L5	Room	Room number, room type (single/double/ward/isolation/procedure/OT/lab), max occupancy, attached bathroom (Y/N), negative pressure (Y/N), HEPA (Y/N)	Room 301 = Single AC room, attached bath, O2+suction
L6	Bed/Station	Bed ID, bed type (standard/ICU/electric/bariatric/crib/bassinet), monitored (Y/N), equipment attached, current patient, cleaning status	Bed ICU-A-05 = Electric ICU bed, ventilator port, monitor arm, IV pole

2.2 Infrastructure Mapping per Location

Each physical location has infrastructure attributes that determine what clinical activities can happen there:

Infrastructure	Configurable Per Location	Values	Impact on Workflows
Medical Gas	O2, N2O, Medical Air, Vacuum/Suction, CO2	Available/Not Available per gas per bed	If O2 not available at bed, system warns during ICU admission. Portable O2 auto-ordered.
Power Backup	UPS tier, DG zone, solar backup	Tier 1 (< 10ms switchover, for OT/ICU), Tier 2 (< 15s, for wards), Tier 3 (no backup)	Equipment requiring uninterrupted power cannot be placed in Tier 3 zones. System warns.
Network	VLAN assignment, WiFi zone, wired ports	Clinical VLAN, Medical Device VLAN, Guest WiFi, Admin VLAN	ABAC uses network zone for access control. Medical device data only flows on device VLAN.
Nurse Call	Call point ID, linked nurse station, escalation chain	Button → Room Nurse → Charge Nurse → Supervisor	Response time SLA tracking per call point.
Intercom	Extension number, linked department, auto-dial for emergency	Ext 3001 = ICU Nurse Station. Code Blue auto-dials: 3001, 1001 (ER), 9999 (switchboard)	Code Blue workflow uses intercom mapping.

CCTV	Camera ID, view area, recording server, retention days	Camera C-ICU-01, 30-day retention, motion detection	Security module links camera to physical zone for incident investigation.
Fire Safety	Fire zone, extinguisher locations, sprinkler zone, smoke detector zone, nearest fire exit	Zone F3A, 2 extinguishers (CO2), sprinkler active, exit via Stairway B	Fire safety module maps equipment to zone. Mock drill routes auto-generated.

2.3 Bed State Machine

Every bed has a state that drives the entire hospital flow. The bed state machine is configurable but defaults to:

State	Trigger	Auto-Actions
Vacant-Clean	Housekeeping marks cleaning complete	Bed appears in available pool. Bed manager can allocate.
Vacant-Dirty	Patient discharged OR transferred out	Housekeeping notified. Bed removed from available pool. Cleaning SLA timer starts (default: 60 min).
Reserved	Bed allocated to incoming patient (admission/transfer)	Bed removed from available pool. Reserved for 2 hours (configurable). If patient doesn't arrive → auto-release back to available.
Occupied	Patient admitted/transferred in	Patient-bed link active. Department dashboard shows occupancy. Equipment at bed linked to patient.
Occupied-Transfer-Pending	Transfer order placed, awaiting receiving ward acceptance	Both source and destination bed managers see pending transfer. If rejected, patient stays.
Maintenance	Equipment repair, infrastructure issue, deep cleaning	Bed removed from all pools. Facilities team work order auto-created. Expected return date tracked.
Blocked	Infection control hold, structural issue, legal hold	Bed completely unavailable. Only Infection Control Officer or Facilities Manager can unblock. Reason mandatory.

State transitions are configurable. Hospital can add custom states (e.g., 'Reserved-for-Post-Op' in hospitals where OT patients have guaranteed ICU beds).

3. Layer 2: Department & Service Configuration

3.1 Department Hierarchy Model

Every hospital structures departments differently. The system supports unlimited nesting with a tree model:

Hospital Type	Example Structure	Depth	Why It Differs
Large Medical College	Medicine → General Medicine → Cardiology Unit → Interventional Cardiology Sub-Unit	4 levels	NMC requires separate departments for recognition. Each sub-specialty has own faculty, beds, OPD.
Multi-Specialty Private	Medicine (covers all sub-specialties) → Cardiology Clinic (just OPD, shared beds)	2 levels	Private hospitals optimize for efficiency, not NMC norms. Cardiologist shares medicine ward.
District Hospital	Medicine Department (1 HOD, MOs cover everything)	1 level	No sub-specialization. One department handles all medical patients.
Standalone Eye Hospital	Cornea Unit, Retina Unit, Glaucoma Unit, Oculoplasty Unit, Pediatric Ophthalmology	1 level (5 parallel units)	Flat structure, all under Ophthalmology. No parent 'Medicine' department.

3.2 Department Configuration Attributes

Attribute	Description & Impact
Department Code + Name	Unique code (MED, SUR, PED, etc.) used across system for referencing. Name displayed on all interfaces.
Parent Department	NULL for top-level. Links to parent for hierarchy. Determines ABAC access inheritance.
Department Type	Clinical / Pre-Clinical / Para-Clinical / Administrative / Support / Academic. Determines which modules are relevant.
OPD Configuration	OPD rooms assigned, consultation slots per day, slot duration (configurable per doctor), walk-in vs appointment ratio, maximum queue size.
IPD Configuration	Wards assigned, bed count, bed types, admission criteria, average LOS (for bed planning), discharge time target.
OT Allocation	Which OTs this department can use, priority days/slots, emergency OT access rules.
Investigation Ordering Rights	Which lab tests and imaging studies this department's doctors can order directly vs requiring referral.
Formulary Access	Which medications this department can prescribe. Sub-specialty formularies possible (e.g., Cardiology has access to cardiac drugs that General Medicine doesn't).
Consultation Fee Structure	OPD fee per doctor designation (Professor/Associate/Assistant), follow-up fee, procedure fees. Different for each department.
NMC Requirements (Medical College)	Minimum beds, minimum OPD/day, minimum faculty (Professor + AP + Asst Prof), minimum procedures per year. System tracks and alerts.
Equipment Assignment	Which equipment is assigned to this department. Feeds into CMMS module for department-wise maintenance tracking.
Quality Indicators	Which NABH quality indicators this department reports on. Auto-data-collection configured per indicator.

3.3 Service Catalog

Every clinical and non-clinical service the hospital offers is a configurable entry in the Service Catalog:

Attribute	Description
Service Code + Name	Unique code (e.g., SRG-APP-001 = Appendectomy). Links to ICD-10-PCS / CPT for coding.
Service Type	Consultation / Procedure / Investigation / Surgery / Therapy / Nursing / Support / Administrative
Department	Which department offers this service. Can be multi-department (e.g., ECG available from Cardiology and ER).
Prerequisites	What must be done before this service: Consent required (Y/N + type), Pre-assessment required (Y/N + type), Investigation required (e.g., blood group before surgery), Fasting required.
Resource Requirements	Room type needed, equipment needed, staff needed (by role and count), estimated duration, consumables list.
Billing	Base price, tax applicability, package linkage, insurance category code, PMJAY HBP package mapping, CGHS rate.
Workflow Link	Which workflow template is triggered when this service is ordered. E.g., ordering 'CT Scan' triggers WF-RAD-001 (Imaging Order to Report).
NABH/NMC Mapping	Which quality indicator or NMC clinical material count this service contributes to. Auto-counted.
Active/Inactive	Hospital can activate or deactivate services without deleting. Inactive services cannot be ordered but historical data preserved.

4. Layer 3: Workflow Engine Architecture

4.1 Workflow Engine Design

The Workflow Engine is the brain of the system. Every process — from patient registration to equipment condemnation — is a configurable workflow template composed of steps, conditions, actions, and transitions.

4.1.1 Workflow Building Blocks

Block Type	Properties	Example
Start Event	Trigger type: Manual (user initiates), Scheduled (time-based), Event (system event triggers), API (external system triggers)	Patient arrives at reception → Manual trigger. Lab result enters critical range → Event trigger. Daily 6 AM → Scheduled trigger for pending PM list.
Task (Human)	Actor (role), Form (data entry), SLA (time limit), Escalation (if SLA breached), Instructions, Required/Optional, Can Skip (Y/N + who can skip)	'Record Vitals': Actor=Nurse, Form=Vitals Form, SLA=10 min after patient checked in, Escalation=Charge Nurse
Task (System)	Auto-action: Calculate, Generate, Send, Create, Update, Check, Transform	Calculate BMI from height+weight. Generate UHID. Send SMS. Create lab order in LIS. Update bed status.
Decision Gateway	Condition: Attribute-based (same syntax as ABAC rules). Exclusive (XOR — one path), Parallel (AND — all paths), Inclusive (OR — one or more paths)	IF patient.age < 18 → Pediatric path. IF patient.mlc_flag = true → MLC path AND standard path (parallel).
Approval Step	Approver (role/specific user), Approval type (single/multi-level/committee), Timeout action (auto-approve/auto-reject/escalate)	Discount > 15% → Finance Head approval. Timeout 4 hours → escalate to CFO.
Sub-Workflow	Embedded workflow call. Main workflow waits for sub-workflow completion or continues in parallel.	Admission workflow calls 'Insurance Verification' sub-workflow in parallel while clinical admission proceeds.
Timer Event	Wait for duration, wait until specific time, deadline	Wait 6 hours then check if discharge summary is signed. If not → alert HOD.
Notification	Channel (SMS/WhatsApp/Email/In-App/Push), Recipient (role/specific/patient/family), Template, Priority	Critical lab value → SMS to ordering doctor (immediate) + In-app to ward nurse (immediate) + Email to HOD (within 1 hour).
End Event	Workflow completion, Workflow termination (error), Workflow cancellation (user-initiated)	Patient discharged → End. Patient DAMA → Cancellation end (triggers different post-actions).

4.1.2 Workflow Template vs Instance

Template: The configurable blueprint (e.g., 'OPD Consultation Workflow'). Created during hospital setup. Can be modified anytime by authorized admin. Versioned — old instances continue on old version, new instances use new version.

Instance: A running copy of the template for a specific patient/case/transaction (e.g., 'OPD Consultation for Patient UHID-12345 on 15-Mar-2026'). Tracks current step, completion status, timestamps, actors, data captured. Cannot be structurally modified — only template can be changed.

Audit Trail: Every step completion in every instance is logged: Who did it, when, from which device/location, what data was entered, how long it took, whether SLA was met.

4.2 Default Workflow Templates (120+)

The system ships with 120+ pre-built workflow templates organized by module. Hospital can use as-is, customize, or build from scratch. Below is the complete catalog with every step documented.

4.3 Patient Registration & Front Office Workflows

WF-REG-001: New Patient Registration

#	Step	Actor	Type	Details / Auto-Actions
1	Patient arrives	Patient	Start (Manual)	Trigger: Receptionist initiates OR patient self-service kiosk
2	Check for existing record	System	System Task	Auto-search by: Phone number → Aadhaar → Name+DOB. If match found → show existing records → receptionist confirms identity → skip to step 6.
3	Capture demographics	Receptionist	Human Task	Form: Name, DOB/Age, Gender, Phone, Address, ID proof, Photo capture. CONFIGURABLE: Hospital can add/remove fields. Aadhaar: Optional but prompted.
4	ABHA ID check	System	Decision	IF patient has ABHA → verify via ABDM API → link. IF no ABHA → offer creation (optional). IF hospital mandates ABHA → route to ABHA creation sub-workflow.
5	Generate UHID	System	System Task	Auto-generate unique hospital ID. Format configurable: [HospitalCode]-[Year]-[Sequence] or [Sequence] only. QR code generated.
6	Patient category assignment	Receptionist	Human Task	Category: General / Private / Insurance / PMJAY / CGHS / Staff / VIP / MLC. CONFIGURABLE: Each hospital defines its categories.
7	Print patient card	System	System Task	Auto-print patient card with UHID, QR code, name. CONFIGURABLE: Card format, printer, whether to print or just QR on phone.
8	Route to destination	System/Receptionist	System Task + Decision	IF OPD visit → Route to department queue. IF ER → Route to triage. IF investigation only → Route to lab/radiology. IF admission → Route to admission workflow.

Configuration Points: Fields in registration form (20+ configurable). ID proof types accepted. Photo mandatory/optional. ABHA creation mandatory/optional. UHID format. Patient categories. Card printing enabled/disabled. Self-service kiosk flow enabled/disabled.

WF-REG-002: Emergency Registration (Fast-Track)

#	Step	Actor	Type	Details / Auto-Actions
1	Patient arrives at ER	Triage Nurse / Security	Start	Trigger: Ambulance arrival OR walk-in at ER gate. Time-stamped to second.
2	Minimal registration	ER Registrar	Human Task	MINIMUM DATA: Name (or 'Unknown Male/Female'), estimated age, gender, brought by whom (name+phone). SLA: 2 minutes maximum. Everything else can be filled later.
3	Temporary UHID	System	System Task	Auto-generate temp UHID if patient is unknown. Format: ER-[Date]-[Sequence]. Can be merged with permanent UHID later.
4	MLC check	ER Doctor / Nurse	Decision	IF MLC (accident, assault, poisoning, burns, sexual

				assault, brought by police, unknown, unconscious): Flag as MLC → auto-trigger WF-MLC-001 (Police Notification + MLC Register) in parallel.
5	Triage	Triage Nurse	Human Task	Manchester/ESI triage scoring. CONFIGURABLE: Which triage system. Category assignment: Resuscitation (Red) / Emergency (Orange) / Urgent (Yellow) / Standard (Green) / Non-Urgent (Blue).
6	Route to area	System	System Task	Auto-route based on triage: Red → Resuscitation Bay (bed auto-assigned). Orange → Acute Care. Yellow → Assessment Area. Green/Blue → Waiting Area / Minor Procedure Room.
7	Complete registration (deferred)	Receptionist / Registration Staff	Human Task	Full registration completed when patient is stable. Family/attendant provides details. Temp UHID merged with permanent. SLA: 24 hours from admission.

WF-REG-003: ABHA ID Creation

#	Step	Actor	Type	Details / Auto-Actions
1	Initiate ABHA creation	Receptionist	Start	Trigger: During registration if patient has no ABHA. OR patient requests ABHA from patient portal.
2	Aadhaar verification	System	System Task	ABDM API call: Aadhaar number → OTP to patient phone → OTP verification. IF Aadhaar not available → Driving License / PAN based creation.
3	Generate ABHA	System	System Task	ABDM API: Create health ID → Receive ABHA number (14-digit) + ABHA address (@abdm). Store in patient master.
4	Consent capture	Patient	Human Task	ABDM consent framework: Patient agrees to link health records. Granular consent: Which records, which providers, validity period. CONFIGURABLE: Default consent template per hospital.
5	Link to hospital records	System	System Task	Auto-link ABHA to UHID. All future clinical records FHIR-tagged for ABDM sharing.

4.4 OPD Workflows

WF-OPD-001: Standard OPD Consultation

#	Step	Actor	Type	Details / Auto-Actions
1	Patient check-in	Patient/Receptionist	Start	Token generated. Queue position assigned. SMS/display notification. CONFIGURABLE: Token format, queue algorithm (FIFO / priority / appointment-first).
2	Vital signs recording	OPD Nurse	Human Task	Form: BP, Pulse, Temp, SpO2, Weight, Height, BMI (auto-calc), Pain Score, Blood Sugar (if diabetic). CONFIGURABLE: Which vitals are mandatory per department.
3	Triage assessment (if configured)	Nurse	Decision	CONFIGURABLE: Some OPDs do nurse triage (e.g., ER-adjacent clinics). IF enabled → Nurse triage → priority re-assignment.
4	Preliminary history (teaching hospital)	Intern	Human Task	CONFIGURABLE: Only in teaching hospitals. Intern records chief complaint, history. Marked as 'Intern Entry — Requires Verification'. Auto-logged in intern logbook.
5	Resident examination (teaching hospital)	Resident	Human Task	CONFIGURABLE: Resident examines, documents findings, preliminary diagnosis. Senior Resident may prescribe within formulary. Case auto-logged in PG logbook.
6	Consultant review	Consultant	Human Task	Reviews intern/resident entries (if teaching) OR does full consultation (if non-teaching). Captures: Diagnosis (ICD-10), Treatment plan, Prescriptions, Investigation orders, Referrals, Follow-up date. CONFIGURABLE: Specialty-specific casesheet form (cardiology form differs from ENT form — Layer 4).
7	Order generation	System	System Task	Based on consultant decisions: Prescription → WF-PHR-001 (Pharmacy Dispensing). Lab orders → WF-LAB-001 (Lab Order-to-Report). Imaging → WF-RAD-001 (Imaging Order-to-Report). Referral → WF-OPD-003 (Referral). Admission → WF-IPD-001 (Admission).
8	Billing	System + Billing Exec	System + Human	Auto-generate consultation bill. CONFIGURABLE: Auto-charge (add to patient account) vs manual billing. IF insurance → pre-auth check. IF PMJAY → scheme rate applied.
9	Pharmacy dispensing	Pharmacist	Sub-Workflow	WF-PHR-001 triggered. Patient proceeds to pharmacy counter. See pharmacy workflows.
10	Checkout	System	End	Patient exits. Follow-up appointment auto-suggested based on doctor's follow-up date. SMS/WhatsApp reminder auto-scheduled.

Hospital Variations — All Configurable:

Teaching Hospital: Steps 4+5 exist (intern + resident before consultant). Non-Teaching: Direct to consultant. Solo Practice: Doctor does everything in one step. Tele-OPD: Video consultation replaces physical steps, e-prescription generated.

WF-OPD-002: Follow-Up Visit

#	Step	Actor	Type	Details / Auto-Actions
1	Patient check-in	Patient/Receptionist	Start	System auto-detects follow-up (same doctor, within follow-up window). CONFIGURABLE: Follow-up window (7 days / 15 days / 30 days). Reduced fee or free follow-up within window.
2	Investigation review	System	System Task	Auto-fetch: Any lab/imaging results since last visit. Display to doctor with previous visit summary side-by-side. Highlight abnormal results.
3	Consultation	Consultant	Human Task	Doctor reviews results, reassesses, modifies treatment. Fewer fields required vs new visit. CONFIGURABLE: Follow-up form (shorter than new patient form).
4	Updated orders	System	System Task	New/modified prescriptions, investigations, referrals generated. Workflow branches as in WF-OPD-001 step 7.
5	Next follow-up / Discharge	Doctor	Decision	IF continue treatment → Schedule next follow-up. IF cured → Discharge from OPD care. IF admit → Trigger WF-IPD-001. IF refer → Trigger WF-OPD-003.

4.5 IPD Workflows

WF-IPD-001: Admission Workflow

#	Step	Actor	Type	Details / Auto-Actions
1	Admission request	Doctor	Start	Source: OPD (consultant recommends), ER (emergency), OT (post-surgical), Transfer from other hospital. Captures: Provisional diagnosis, admission type (emergency/elective), bed type requested, expected LOS.
2	Bed allocation	Bed Manager / System	Human/System	System shows available beds matching criteria (department, bed type, gender). CONFIGURABLE: Auto-assign (first available) vs manual selection. IF no bed available → waitlist with priority scoring.
3	Insurance verification	Insurance Coordinator	Sub-Workflow (Parallel)	Triggered in parallel with clinical admission. IF insured → WF-BIL-002 (Pre-auth). IF PMJAY → WF-BIL-003 (Scheme verification). IF cash → skip. Does NOT block clinical admission.
4	Admission documentation	Nurse	Human Task	Nursing admission assessment: Fall risk (Morse scale), pressure ulcer risk (Braden), nutritional screening (MUST), allergy documentation, valuables inventory, patient belongings list. CONFIGURABLE: Which assessments are mandatory per ward type.
5	Consent capture	Nurse / Doctor	Human Task	General admission consent, treatment consent. CONFIGURABLE: Which consents auto-triggered per admission type. System blocks further progress until mandatory consents are signed.
6	Initial orders	Admitting Doctor	Human Task	Diet order (→ Dietary dept), medication orders (→ Pharmacy), investigation orders (→ Lab/Radiology), nursing orders (monitoring frequency, I/O charting, special instructions). Each order triggers respective module workflow.
7	Notification cascade	System	System Task	Auto-notify: Nursing station (bed assignment), Kitchen (diet order), Pharmacy (medication orders), Housekeeping (bed preparation if needed), Ward secretary. CONFIGURABLE: Who gets notified, via which channel.
8	Academic logging (teaching)	System	System Task	IF teaching hospital AND admitting team includes PG/intern → auto-log in academic module. Case type mapped to PG logbook categories.

WF-IPD-002: Daily Ward Round

#	Step	Actor	Type	Details / Auto-Actions
1	Round initiation	Consultant/Resident	Start	CONFIGURABLE: Round time (morning 8-10 AM default). System prepares round list: Patient list with summary cards showing last 24hr vitals trend, new investigation results, pending results, nurse notes, I/O chart summary.
2	Patient review	Doctor	Human Task	Per patient: Review vitals trend (auto-charted),

				investigation results (new highlighted), nursing notes (flagged items), medication chart (compliance). CONFIGURABLE: Which data is auto-displayed per specialty (ICU shows ventilator parameters, Medicine shows I/O balance).
3	Progress note	Doctor	Human Task	CONFIGURABLE: SOAP format / narrative / problem-oriented. Teaching hospital: Intern writes, resident reviews, consultant approves. Non-teaching: Doctor writes directly.
4	Order modification	Doctor	Human Task	New/modified: Medications, investigations, diet, activity level, monitoring frequency. Each change → auto-routed to respective department.
5	Discharge planning check	Doctor	Decision	IF patient ready for discharge → Trigger WF-IPD-003 (Discharge). IF transfer needed → Trigger WF-IPD-004 (Transfer). IF surgery needed → Trigger WF-OT-001 (Surgical Scheduling). CONFIGURABLE: Discharge readiness criteria (automated checklist).

WF-IPD-003: Discharge Workflow

#	Step	Actor	Type	Details / Auto-Actions
1	Discharge order	Consultant	Start	Discharge type: Normal / DAMA / Transfer / Death / Abscond. CONFIGURABLE: Who can initiate each type. DAMA requires specific consent form + risk acknowledgment.
2	Discharge summary preparation	Resident/Doctor	Human Task	CONFIGURABLE: Auto-populated from admission data, diagnosis, procedures, investigation results, medication at discharge. Doctor adds: Condition at discharge, follow-up instructions, warning signs. Format configurable per hospital.
3	Medication reconciliation	Pharmacist/Nurse	Human Task	Compare: Admission medications vs discharge medications. Flag: New additions, modifications, discontinued. Ensure patient understands changes. CONFIGURABLE: Mandatory for certain departments (e.g., Cardiology, Geriatrics).
4	Nursing discharge assessment	Nurse	Human Task	Wound care instructions, mobility status, feeding instructions, follow-up appointments, home care needs assessment. CONFIGURABLE: Specialty-specific discharge checklist.
5	Dietician instructions (if applicable)	Dietician	Human Task	CONFIGURABLE: Auto-triggered for: Diabetic patients, post-surgical patients, renal patients, cardiac patients. Diet plan printout in patient's language.
6	Final billing	Billing Team	Human Task	System auto-generates final bill from all charges accumulated during stay. CONFIGURABLE: Bill review mandatory (some hospitals allow auto-finalization for package cases). Discount workflow if applicable.
7	Insurance settlement	Insurance Coordinator	Sub-Workflow	IF insured → Final claim submission. IF PMJAY → Package claim processing. IF TPA → Claim

				documents compilation. CONFIGURABLE: Runs parallel with patient discharge or blocks discharge until settlement.
8	Patient education & handover	Nurse	Human Task	Medications explained, warning signs discussed, follow-up appointment confirmed, emergency contact numbers provided. CONFIGURABLE: Patient education material (auto-selected by diagnosis, generated in patient's language).
9	Clearance verification	System	System Task	Auto-check: All consents documented? Discharge summary signed? Billing complete (or payment plan agreed)? Valuables returned? No pending investigation results? CONFIGURABLE: Which checks are blocking vs warning.
10	Bed release	System	System Task	Bed status → Vacant-Dirty. Housekeeping notified. Discharge time stamped. ABHA notification sent (health record updated). SMS to patient: 'Thank you, rate your experience' link.

4.6 Emergency Department Workflows

WF-ER-001: Emergency Triage & Stabilization

#	Step	Actor	Type	Details / Auto-Actions
1	Arrival	Security / Nurse	Start	Patient arrival logged. Source: Walk-in, ambulance (EMS data auto-imported if available), inter-hospital transfer (referral note auto-scanned). TIME STAMP is medico-legally critical.
2	Quick registration	ER Registrar	Human Task	Parallel with triage (does NOT delay clinical care). See WF-REG-002.
3	Triage assessment	Triage Nurse	Human Task	CONFIGURABLE: Manchester Triage System (5 levels) OR Emergency Severity Index (5 levels) OR hospital-custom. Form fields per triage system. Auto-assigns urgency category.
4	Area assignment	System	System Task	Red → Resuscitation Bay (bed auto-assigned, crash cart verified, team alerted). Orange → Acute area. Yellow/Green/Blue → appropriate area. CONFIGURABLE: Area mapping per triage category per hospital layout.
5	ER Doctor assessment	CMO/ER Doctor	Human Task	Primary survey (ABCDE), secondary survey. CONFIGURABLE: Trauma vs medical vs pediatric vs obstetric assessment forms. MLC auto-detected from presentation (RTA, assault, poisoning, burns, unknown patient).
6	STEMI / Stroke pathway	System	Decision	CONFIGURABLE: If chest pain → auto-trigger STEMI pathway (WF-CATH-001, Door-to-Balloon timer starts). If stroke symptoms → auto-trigger Stroke pathway (Door-to-Needle timer starts). Timer prominently displayed, auto-escalates if SLA breached.
7	Stabilization & disposition	ER Doctor	Human Task + Decision	After stabilization: Admit (→ WF-IPD-001), Transfer to OT (→ WF-OT-001), Observe in ER (→ timer set for reassessment), Discharge from ER (→ WF-ER-003), Transfer to other hospital (→ WF-ER-004), Death (→ WF-MORT-001). CONFIGURABLE: Disposition options per hospital capability.

WF-ER-002: Mass Casualty Incident (MCI)

#	Step	Actor	Type	Details / Auto-Actions
1	MCI declaration	Medical Superintendent / CMO	Start	CONFIGURABLE: Who can declare MCI (role-based). Requires confirmation (prevent accidental activation).
2	System-wide mode change	System	System Task	AUTOMATIC: Access controls expanded (all clinical staff → ER-level access). Registration simplified (triage tag = temporary ID). Billing deferred. All elective surgeries paused. OT reserved for MCI. Non-essential departments notified. Staff recall initiated (off-duty staff auto-SMS).

3	Triage (START/JumpSTART)	Triage Teams	Human Task	CONFIGURABLE: START (adult) / JumpSTART (pediatric) / hospital custom. Color-coded tags: Red (immediate), Yellow (delayed), Green (minor), Black (dead/expectant). Rapid tagging — system accepts minimal data.
4	Treatment area management	MCI Coordinator	Human Task	Red → Resuscitation area (multiple bays). Yellow → Observation area. Green → Minor treatment + discharge. Black → Mortuary holding. CONFIGURABLE: Area designation per hospital physical layout.
5	Resource tracking	System	System Task	Real-time dashboard: Patients by category, beds available, OT status, blood bank inventory, ventilator availability, staff deployment. Auto-requests: Blood bank → alert for mass transfusion readiness. Pharmacy → emergency medication stock check.
6	Communication	System	System Task	Auto-notifications: District Collector, police, nearby hospitals (for overflow), media spokesperson (if configured), ambulance services. CONFIGURABLE: Communication tree per local disaster management plan.
7	MCI deactivation	Medical Superintendent	End	Explicit deactivation. System reverts to normal mode. Post-event: All MCI patient records compiled. Government reporting auto-generated. Debrief scheduled.

4.7 Operation Theatre Workflows

WF-OT-001: Surgical Scheduling & Clearance

#	Step	Actor	Type	Details / Auto-Actions
1	Surgery request	Surgeon	Start	Captures: Patient, procedure(s), urgency (elective/urgent/emergency), preferred date/time, OT preference, anesthesia type, expected duration, special equipment needed, blood requirement.
2	OT slot check	System	System Task	Auto-check: OT availability for requested date/slot. Equipment availability. Anesthesiologist availability. Conflict detection. IF emergency → bypass scheduling, assign first available OT.
3	Pre-anesthetic assessment	Anesthesiologist	Human Task	PAC (Pre-Anesthetic Checkup). Mandatory GATE — surgery CANNOT proceed without PAC clearance. Form: Airway assessment (Mallampati), cardiac risk (ASA class), medications review, allergy review, anesthesia plan, risk discussion. CONFIGURABLE: PAC form per hospital + additional fields per anesthesiologist preference.
4	Pre-operative investigations	System	Decision	CONFIGURABLE: Per procedure type: What investigations are mandatory. E.g., Major surgery: CBC, RFT, LFT, coagulation, blood grouping, ECG, chest X-ray. Minor surgery: CBC, blood sugar. System checks if results are available and within validity (e.g., CBC within 7 days). IF missing → auto-order. IF expired → re-order.
5	Consent verification	System	System Task	System verifies: Surgical consent signed (procedure-specific), anesthesia consent signed, blood transfusion consent (if applicable), high-risk consent (if applicable). BLOCKS if any missing.
6	OT scheduling confirmation	OT Coordinator	Human Task + Approval	Final OT list published. CONFIGURABLE: List publication time (previous day 6 PM default). Distribution: Surgeon, anesthesiologist, OT nurses, CSSD, blood bank, ward.
7	Pre-op preparation	Ward Nurse	Human Task	Checklist: Fasting status, site marking (correct side), skin preparation, pre-medication given, blood products available, consent forms attached, investigation reports attached. CONFIGURABLE: Checklist items per procedure type.

WF-OT-002: Intra-Operative — WHO Surgical Safety Checklist

#	Step	Actor	Type	Details / Auto-Actions
1	Sign In (before anesthesia)	OT Nurse + Anesthesiologist	Human Task	MANDATORY GATE. Confirms: Patient identity (barcode scan), procedure + site, consent, site marking, allergy check, airway assessment, blood loss risk + blood available, pulse oximeter functioning. ALL items must be checked — system blocks progression if any unchecked.

2	Time Out (before incision)	Entire Team	Human Task	MANDATORY GATE. Confirms: All team members introduced, patient + procedure + site confirmed, anticipated critical events reviewed, antibiotic prophylaxis given (within 1 hour of incision), imaging displayed. ALL items must be checked.
3	Surgical procedure	Surgeon + Team	Human Task	Operative notes in real-time or post-procedure. Specimen labeling (barcode). Implant logging (barcode scan). Instrument count ongoing. Equipment malfunction documentation if any. Anesthesia record continuous (auto-capture from monitors if connected).
4	Sign Out (before leaving OT)	OT Nurse + Surgeon + Anesthesiologist	Human Task	MANDATORY GATE. Confirms: Procedure name recorded, instrument + swab + needle count correct, specimen labeled, equipment problems documented, post-op care instructions communicated. HARD BLOCK: If instrument count mismatch → workflow halts. Only surgeon can override with documented explanation + X-ray order.
5	Post-op handover	Anesthesiologist → Recovery Nurse	Human Task	SBAR handover: Situation, Background, Assessment, Recommendation. Includes: Pain management plan, monitoring frequency, recovery milestones for discharge from recovery. CONFIGURABLE: Recovery scoring (Aldrete / Modified Aldrete / hospital custom).

4.8 Laboratory Workflows

WF-LAB-001: Investigation Order to Report (Complete Cycle)

#	Step	Actor	Type	Details / Auto-Actions
1	Investigation order	Doctor	Start	Order placed from OPD/IPD/ER. Auto-fields: Patient details, ordering doctor, clinical indication, priority (routine/urgent/STAT). CONFIGURABLE: Order approval required for expensive tests (e.g., genetic tests, advanced panels), order sets per diagnosis (chest pain panel = Troponin + CK-MB + ECG + CBC), department-specific order restrictions.
2	Order validation	System	System Task	Auto-check: Duplicate order (same test within configurable window — default 24hr for routine, 4hr for STAT). Insurance coverage. Patient preparation required (fasting). Sample type required. Tube type required. CONFIGURABLE: Duplicate check windows per test.
3	Sample collection	Phlebotomist	Human Task	Barcode label generated. Phlebotomist scans: Patient wristband (IPD) or UHID card (OPD) → sample tube barcode. Mismatch = HARD BLOCK. Captures: Collection time, collection site, collector ID, sample condition. CONFIGURABLE: Mobile collection (bedside), collection center, outsourced collection.
4	Sample receipt at lab	Lab Technician	Human Task	Scan barcode at lab receipt point. Auto-log: Transit time (collection to receipt — KPI). Rejection check: Hemolyzed, lipemic, icteric, insufficient volume, wrong tube, unlabeled. IF rejected → notify collector + ordering doctor + auto-trigger recollection.
5	Sample processing	Lab Technician	Human Task + System	Section routing (hematology/biochemistry/micro/etc.). Instrument loading (auto via track system or manual). CONFIGURABLE: Which analyzers are primary/backup per test. Interface mapping: Test code → analyzer channel → ASTM/HL7 result mapping.
6	QC check	System	Decision	Before patient results: System checks if daily IQC has been run and passed (Westgard rules). IF QC failed → BLOCK patient result processing → alert QC officer → corrective action required. CONFIGURABLE: QC rules per test, Westgard rule set (1-2s, 1-3s, 2-2s, R-4s, 4-1s, 10x).
7	Result entry / auto-capture	System + Technician	System/Human	Automated: Results flow from analyzer via interface → system captures automatically. Manual: Technician enters results (microscopy, culture). CONFIGURABLE: Which tests are auto-captured vs manual.
8	Auto-validation	System	Decision	CONFIGURABLE auto-validation rules per test: IF result within normal range AND no delta check failure AND QC passed AND not STAT AND patient

				age > 12 AND sample not flagged → auto-validate (no pathologist review needed). IF any criteria fails → route to pathologist.
9	Pathologist validation	Pathologist	Human Task	Reviews flagged results. Can: Approve, reject (rerun), add comments, order reflex testing (e.g., abnormal CBC → add peripheral smear). Digital signature applied. CONFIGURABLE: Which pathologist reviews which section.
10	Critical value alert	System	System Task + Decision	IF result in critical range → BLOCK report release until ordering doctor acknowledges. Notification cascade: 5 min → ordering doctor (SMS + in-app). 15 min → ward nurse. 30 min → department HOD. 60 min → Medical Superintendent. CONFIGURABLE: Critical value ranges per test per hospital.
11	Report release	System	System Task	Report available in HMS. Auto-notify ordering doctor. Patient can view on portal (if enabled). CONFIGURABLE: Report format, normal ranges display, interpretive comments. ABDM: Report auto-linked to ABHA if consented.
12	TAT tracking	System	System Task	Auto-calculate: Order-to-collection, collection-to-receipt, receipt-to-result, result-to-report. Compare against SLA. CONFIGURABLE: SLA per test category (STAT: 1 hr, Routine: 4-6 hr, Culture: 48-72 hr).

4.9 Pharmacy Workflows

WF-PHR-001: Prescription to Dispensing

#	Step	Actor	Type	Details / Auto-Actions
1	Prescription received	System	Start	Auto-received from OPD/IPD/ER. Contains: Drug name (from formulary), dose, frequency, route, duration. System auto-checks at order time: Drug-drug interaction, drug-allergy cross-check, duplicate therapy, renal/hepatic dose adjustment needed, drug-food interaction, drug-lab interaction, LASA (look-alike/sound-alike) warning, pregnancy category warning.
2	Interaction override (if any)	Doctor	Decision	IF interaction detected: Severity levels: Info (auto-proceed with note), Warning (doctor must acknowledge), Critical (doctor must provide clinical justification, pharmacist must verify). CONFIGURABLE: Interaction severity levels, override authority per role.
3	Stock check	System	System Task	Auto-check: Drug availability in dispensing location. IF out of stock → suggest: Generic alternative, therapeutic alternative, nearest sub-store with stock, procurement timeline. CONFIGURABLE: Substitution rules (generic allowed, therapeutic needs doctor approval).
4	Dispensing	Pharmacist	Human Task	Pharmacist scans: Prescription barcode → drug barcode → verifies match. System displays: Drug image (visual verification), dispensing quantity, patient instructions. Controlled substance: Additional register entry. CONFIGURABLE: Barcode verification mandatory/optional, dual verification for high-risk drugs.
5	Patient counseling	Pharmacist	Human Task	CONFIGURABLE: Required for new medications, high-alert medications, complex regimens. Counseling points auto-generated per drug (timing, food interaction, storage, side effects to watch). Documentation of counseling.
6	Dispensing confirmation	System	System Task	Inventory deducted. Billing applied. Dispensing record linked to patient medication history. For IPD: MAR (Medication Administration Record) auto-updated with dispensed medications.

WF-PHR-002: Controlled Substance Management

#	Step	Actor	Type	Details / Auto-Actions
1	Controlled substance prescribed	Doctor	Start	System auto-detects: Schedule H1, Schedule X, NDPS Act drugs. Validates: Prescriber registration number, narcotic license (for NDPS). CONFIGURABLE: Drug schedule classification per country regulation.
2	Dual verification	Two Pharmacists	Human Task	Two pharmacists independently verify: Prescription

				validity, quantity, prescriber authority. Both sign (digital). CONFIGURABLE: Single verification for Schedule H1, dual for Schedule X and NDPS.
3	Register entry	Pharmacist	Human Task	Controlled Substance Register: Date, patient name, prescriber name, drug name, quantity, batch number, balance stock. Auto-running balance calculation. Daily closing balance reconciliation. CONFIGURABLE: Register format per state Drug Control authority.
4	Dispensing with chain-of-custody	Pharmacist	Human Task + System	Drug removed from controlled substance safe (biometric access + dual authorization). Quantity counted, documented. Handed to patient/nurse (signature obtained). CONFIGURABLE: Safe access rules.
5	Administration (IPD)	Nurse	Human Task	For IPD: Nurse administers at bedside. Barcode scan: Patient + drug + dose verification. Waste documentation if partial ampoule used (witnessed destruction of remaining). CONFIGURABLE: Witness requirement per drug type.
6	Reconciliation	Chief Pharmacist	Scheduled Task	Daily: Physical count vs system balance. Discrepancy investigation if mismatch. Monthly: Comprehensive audit. Annual: Drug inspector submission. CONFIGURABLE: Reconciliation frequency per drug schedule.

5. Cross-Module Automation Links (The Nervous System)

This is what transforms 67 disconnected modules into a single intelligent system. When an event occurs in one module, it automatically triggers actions in other modules. These links are the 'nervous system' of the HMS — they eliminate manual coordination, prevent errors, and ensure nothing falls through the cracks.

Below is the COMPLETE catalog of cross-module automation links. Each link is configurable — hospital can enable/disable any link, modify the trigger conditions, or change the target actions.

5.1 Patient Registration → All Modules

#	Trigger Event	Target Module(s)	Automated Action(s)
1	New patient registered	ABDM Integration	IF ABHA creation chosen → API call to ABDM → Create ABHA → Link to UHID. IF existing ABHA → Fetch health records with consent.
2	Patient registered as MLC	Medico-Legal Module, Communication	Auto-create MLC register entry. Auto-notify: Duty doctor, Medical Superintendent, police liaison. Auto-generate police intimation letter (pending signature).
3	Patient flagged as VIP	Security, ABAC Engine	Enhanced privacy mode activated. Record access restricted to assigned care team. Name aliased in displays. All access logged with enhanced audit.
4	Minor patient (age < 18) registered	Consent Module, ABAC	Guardian mandatory flag set. Auto-generate guardian consent requirement for all procedures. Restrict data sharing without guardian authorization.
5	Patient registered with known allergy	All Clinical Modules	Allergy banner displayed on every clinical screen. Auto-check against all future prescriptions, investigations using contrast, dietary items.
6	Returning patient identified	All Clinical Modules	Previous records auto-available: Last visit summary, active medications, pending results, chronic conditions, previous admissions. Doctor sees continuity of care summary.

5.2 OPD → Multiple Modules

#	Trigger Event	Target Module(s)	Automated Action(s)
7	Doctor places lab order	Laboratory (LIS)	Lab order created in LIS worklist. Collection priority assigned. Barcode labels generated. IF fasting required → patient notification with instructions.
8	Doctor places imaging order	Radiology (RIS)	RIS worklist updated. Scheduling suggested based on equipment availability. IF contrast required → check renal function (auto-fetch latest creatinine). PCPNDT Form F auto-generated for OB ultrasound.
9	Doctor writes prescription	Pharmacy	Prescription appears in pharmacy dispensing queue. Stock check auto-run. Drug interaction check already done (at prescribing time). Billing auto-generated.
10	Doctor refers to another department	Referral target department	Referral notification to target department. Appointment auto-suggested. Referral letter auto-generated with current diagnosis, investigation results, medication list. ABAC: Target department gets read access to relevant patient data.
11	Doctor recommends admission	IPD, Bed Management	Admission request created. Bed search initiated. Insurance pre-auth triggered (if applicable). Nursing admission assessment form auto-queued.
12	OPD visit completed	Academic ERP (teaching)	IF intern/PG documented case → auto-log in logbook: Date, patient UHID (anonymized), diagnosis, procedures done, supervisor name. Logbook entry

			pending supervisor verification.
13	Diagnosis entered (ICD-10)	NMC Compliance, Quality	Daily OPD count auto-incremented per department (NMC clinical material tracking). IF notifiable disease → auto-alert to IDSP (Integrated Disease Surveillance). IF cancer → auto-flag for cancer registry.

5.3 IPD → Multiple Modules

#	Trigger Event	Target Module(s)	Automated Action(s)
14	Patient admitted	Bed Management, Dietary, Housekeeping, Nursing, Billing, ABHA	Bed status → Occupied. Diet order to kitchen. Housekeeping: Room setup checklist. Nursing: Admission assessment due. Billing: Admission charges initiated. ABHA: Encounter created.
15	Medication order placed (IPD)	Pharmacy, Nursing (MAR)	Pharmacy: Dispensing request created. Nursing: MAR updated with scheduled administration times. Drug interaction re-checked against all current IPD medications.
16	Investigation results available	Doctor notification, Nursing	Ordering doctor notified (in-app + badge count). Ward nurse notified. IF critical → escalation cascade. IF abnormal → highlighted in patient chart. IF diabetes: glucose result auto-updates insulin sliding scale (if configured).
17	Vital signs abnormal	Doctor alert, ICU (if applicable)	CONFIGURABLE thresholds per department. IF vitals trigger early warning score (NEWS/MEWS) → auto-calculate score. IF score > threshold → alert treating team + ICU outreach. CONFIGURABLE: Rapid Response Team activation threshold.
18	Patient transferred between wards	Bed Management, Nursing, Dietary, Housekeeping, Billing	Source bed → Vacant-Dirty. Target bed → Occupied. Nursing handover enforced (SBAR). Diet order updated for new location. Billing: Room charges adjusted. Equipment: Any patient-specific equipment transferred.
19	Patient discharged	Bed Mgmt, Billing, Pharmacy, ABHA, Follow-up, Feedback	Bed released. Final bill generated. Discharge medications dispensed. ABHA encounter closed. Follow-up appointment auto-scheduled. Patient satisfaction survey auto-sent (24hr delay configurable).
20	Patient death	Mortuary, Billing, MLC (if applicable), NMC, Quality	Mortuary: Body transfer request. Billing: Final bill + death surcharge waiver (if configured). MLC: If MLC → police notification. NMC: Death added to clinical material statistics. Quality: Death review committee auto-notified. Infection Control: If HAI suspected → investigation triggered.
21	Hospital-acquired infection detected	Infection Control, Quality, Pharmacy (Antibiotic Stewardship)	Infection control: HAI surveillance record created (CLABSI/CAUTI/VAP/SSI classification). Quality: Incident report auto-created. Pharmacy: Antibiotic escalation review triggered. Environment: Enhanced cleaning ordered for bed area.

5.4 Emergency → Multiple Modules

#	Trigger Event	Target Module(s)	Automated Action(s)
22	STEMI patient identified (chest pain + ECG)	Cath Lab, Cardiology, Blood Bank, OT	Door-to-Balloon timer starts (visible on dashboards). Cath Lab team alerted (SMS + pager). Cardiology on-call notified. Blood bank: 2 units PRBC reserved. OT: Emergency slot flagged if cath lab needs bailout surgery. Family: Consent briefing initiated by ER nurse.
23	Stroke patient identified	Neurology, Radiology (CT), Pharmacy	Door-to-Needle timer starts. CT Brain STAT ordered (auto-prioritized in radiology worklist). Neurology on-call alerted. Pharmacy: tPA checked for availability. NIHSS scoring form auto-loaded.
24	Trauma patient — massive	Blood Bank, OT, ICU, Lab	MTP activated: Blood bank issues O-negative (2 units immediate, then type-

	transfusion protocol		specific). Lab: Coagulation panel STAT auto-ordered at 0, 30, 60 min intervals. ICU: Bed reserved. OT: Emergency slot activated if surgical intervention needed.
25	Poisoning case	Lab (Toxicology), Poison Information Center	Lab: Toxicology panel auto-ordered. If known poison → specific antidote availability checked in pharmacy. Poison control center protocol auto-loaded. GI decontamination checklist displayed. MLC auto-flagged.
26	Mass casualty declared	ALL MODULES	See WF-ER-002. Hospital-wide mode change. Every module enters MCI mode with expanded access, simplified workflows, deferred billing.

5.5 OT/Surgery → Multiple Modules

#	Trigger Event	Target Module(s)	Automated Action(s)
27	Surgery scheduled	CSSD, Blood Bank, Pharmacy, Anesthesia, Nursing	CSSD: Instrument set requirements for procedure → sterilization scheduled. Blood Bank: Blood reservation if expected blood loss > 500ml. Pharmacy: Pre-op antibiotics timed for administration 30-60 min before incision. Anesthesia: PAC schedule auto-proposed. Nursing: Pre-op checklist auto-generated for ward nurse.
28	WHO Sign In completed	OT Manager, All OT Staff	Surgery officially 'In Progress'. OT occupancy dashboard updated. No other surgery can be scheduled in this OT until Sign Out.
29	Specimen sent from OT	Pathology (Histopathology)	Specimen barcode auto-registered in pathology worklist. Chain of custody: OT nurse → transport → pathology receipt. Fresh frozen section: If intra-operative consultation requested, pathology team auto-alerted for immediate processing.
30	Implant used in surgery	Equipment/CMMS, Billing, Regulatory	Implant barcode scanned: Serial number, lot number, expiry, manufacturer logged against patient. Billing: Implant cost added to bill. CMMS: Implant linked to patient for recall tracking. Regulatory: National implant registry entry (if applicable).
31	Surgery completed (Sign Out done)	ICU/Ward (post-op), Nursing, Academic	Post-op orders auto-queued for receiving unit (ICU or ward). Nursing handover enforced. Anesthesia record finalized. Operative note auto-generates NMC surgical statistics entry. Academic: If PG performed/assisted → logbook entry auto-created.
32	Instrument/swab count mismatch	OT Team, Radiology	HARD ALERT: Surgery cannot close. X-ray ordered (retained foreign body check). Incident report auto-created. If found → retrieval. If not found after X-ray → surgeon documentation mandatory. Never event tracking.

5.6 Laboratory → Multiple Modules

#	Trigger Event	Target Module(s)	Automated Action(s)
33	Critical lab value detected	Ordering Doctor, Ward Nurse, Quality	See WF-LAB-001 step 10. Escalation cascade. Report blocked until acknowledgment. Incident report if acknowledgment delayed beyond SLA. NABH quality indicator automatically tracked.
34	Antibiotic sensitivity report released	Pharmacy (Antibiotic Stewardship), Infection Control	IF current antibiotic is resistant on sensitivity → auto-alert treating doctor: 'Current antibiotic [X] shows resistance. Sensitive alternatives: [Y, Z].' Antibiotic stewardship dashboard updated. IF MDR organism → Infection Control notified for isolation precautions.
35	Blood culture positive	Infection Control, Treating Team	Auto-alert treating team (STAT). Infection control notified for HAI assessment. IF MRSA/VRE/CRE → isolation recommendation auto-

			generated. Antibiotic review prompted.
36	HIV/Hepatitis positive (new diagnosis)	Counseling, Infection Control, Quality	Counseling appointment auto-scheduled. Infection control: Exposure assessment if needle stick involved. Quality: NABL reporting. Patient notification per protocol (NOT via SMS — in-person counseling mandatory). Enhanced data privacy triggered.
37	Lab QC failure (Westgard rule violated)	Lab Director, QC Officer	Patient sample processing BLOCKED for affected test/analyzer. QC officer notified immediately. Corrective action workflow initiated. Affected results (if any already released) flagged for review. Patients potentially affected identified.
38	Lab equipment breakdown	CMMS/Equipment Module	Auto-create work order in CMMS. Backup equipment workflow activated. Estimated repair time requested. Affected test routing changed to backup analyzer (if configured). Affected pending orders re-queued to backup.

5.7 Pharmacy → Multiple Modules

#	Trigger Event	Target Module(s)	Automated Action(s)
39	Drug stock below reorder level	Materials/Inventory, Purchase	Auto-generate purchase indent. Vendor auto-selected from approved vendor list (best price / shortest delivery). IF critical drug (life-saving) → emergency procurement workflow with Medical Superintendent notification.
40	Drug near expiry (< 90 days)	Pharmacy, Finance	Flagged in dispensing system (FEFO enforcement ensures dispensed first). IF < 30 days and stock high → vendor return initiated. Financial impact calculated. Write-off request if not returnable.
41	Adverse Drug Reaction reported	Quality (Incident Report), Regulatory (PvPI/CDSCO)	Incident report auto-created. PvPI (Pharmacovigilance Programme of India) ADR form auto-populated from patient and drug data. Submission to CDSCO auto-triggered for serious ADRs. Drug flagged in patient's allergy/adverse reaction list permanently.
42	Medication error detected	Quality, Nursing, Treating Doctor	Incident report auto-created. Treating doctor notified for patient assessment. Nursing: If administration error → observation protocol initiated. Root cause analysis triggered if severity ≥ moderate.

5.8 Equipment/CMMS → Multiple Modules

#	Trigger Event	Target Module(s)	Automated Action(s)
43	Life-support equipment breakdown (criticality 5)	Clinical Department, Admin, Vendor, Bed Management	Clinical: Ward/ICU nurse alerted immediately + backup equipment location shown. Admin: Medical Superintendent notified. Vendor: If under AMC → auto-service request sent. Bed Management: Bed marked as 'partially functional' (has patient but critical equipment down). CMMS: SLA timer starts (2-hour resolution target).
44	Calibration due (< 7 days)	Clinical Department, BME Team, Quality	Warning displayed when equipment accessed/used. BME: Calendar reminder for calibration. Quality: Dashboard shows compliance percentage. IF expired without calibration → hard block on clinical use for Class C/D equipment.
45	PM (Preventive Maintenance) due	BME Team, Department	PM work order auto-created per schedule. BME technician assigned (based on skill/location). Department head notified if PM requires equipment downtime during clinical hours. After PM: Return-to-service workflow requires BME sign-off.

46	Equipment condemned	Finance (Write-off), Inventory (De-inventory), Quality	Finance: Asset write-off processed. Depreciation stopped. Inventory: Equipment removed from active inventory. Physical disposal tracked (e-waste for electronic, biomedical waste rules for contaminated). Replacement need flagged in capital budget planning.
47	New equipment commissioned	HR (Training), CMMS, Finance, Regulatory	HR: Training requirement created for all designated operators. CMMS: PM schedule activated, calibration schedule activated. Finance: Asset added to register, depreciation starts. Regulatory: If radiation equipment → AERB notification. ABAC: Equipment-specific training certification tracking activated.

5.9 Billing → Multiple Modules

#	Trigger Event	Target Module(s)	Automated Action(s)
48	Bill generated for IPD discharge	Insurance, Finance, Patient Communication	IF insurance → claim auto-compiled with supporting documents. Finance: Revenue entry in accounting system (Tally/SAP integration). Patient: Bill summary sent via SMS/WhatsApp. IF PMJAY → claim auto-submitted to scheme portal.
49	Insurance pre-auth rejected	Treating Doctor, Patient Relations, Billing	Doctor notified (may need to change procedure or provide additional justification). Patient relations: Family counseled about self-pay option. Billing: Bill switched from insurance to self-pay track. Appeal workflow available.
50	Payment received	Finance, Patient Communication	Finance: Receipt entry, cash/bank reconciliation. Patient: Receipt sent digitally. IF advance received → patient balance dashboard updated. IF final payment → all holds released (record access for MRD, etc.).

5.10 Academic ERP → HMS & Admin Modules

#	Trigger Event	Target Module(s)	Automated Action(s)
51	PG student clinical posting starts	HMS (ABAC), OPD/IPD, Logbook	ABAC: Student gets access to assigned department's clinical data (within their posting). OPD/IPD: Student appears in department's active team list. Logbook: Case logging activated for the department.
52	Intern rotation changes	ABAC, Department Teams, Logbook	Previous department access revoked. New department access granted. New supervisor assigned. Logbook sections updated. Attendance tracking switched to new department.
53	PG performs/assists procedure (logged in OT)	PG Logbook, Academic Assessment	Procedure auto-logged: Date, patient (anonymized UHID), procedure name, role (primary/first assistant/second assistant), supervisor, complications if any. Counts toward NMC minimum procedure requirements per specialty.
54	NMC faculty vacancy created	HR, Academic Admin, NMC Compliance	Vacancy impacts faculty-student ratio calculation. NMC compliance dashboard shows RED if ratio falls below minimum. Dean alerted. Recruitment workflow initiated in HR.
55	AEBAS attendance synced to NMC	NMC Portal, HR	Daily faculty attendance auto-synced via AEBAS API. Discrepancies flagged (biometric present but no clinical/teaching activity). NMC dashboard shows real-time compliance.

5.11 Infrastructure → Clinical & Admin

#	Trigger Event	Target Module(s)	Automated Action(s)
56	O2 supply pressure drop (MGPS)	ICU/OT/ER (Clinical), Medical Superintendent, MGPS Team	Clinical: O2 supply status shown on bedside displays (green → yellow → red). Ventilator-dependent patients auto-identified. Portable O2 cylinders located on floor map. Medical Superintendent: Immediate alert. MGPS: Alarm acknowledged, switchover to backup initiated.
57	Power failure (DG not started within 15 seconds)	ICU/OT/NICU, IT, Medical Superintendent, Facilities	Critical areas: Battery backup status displayed. Equipment running on UPS shown. IT: Server room status. Facilities: Emergency power restoration protocol. IF DG fails completely → disaster protocol consideration.
58	Water quality failure (dialysis RO)	Dialysis Unit, Nephrologist, Facilities	Dialysis: IMMEDIATE HALT of all active dialysis sessions. Nephrologist: Assessment of patients mid-session. Facilities: RO plant troubleshooting. Quality: Incident report auto-created.
59	Fire alarm triggered	All Departments (Code Red), Security, Fire Safety	Code Red workflow activated. Affected zone identified. Evacuation plan displayed on nurse stations. Patient categorization for evacuation. Fire safety officer notified. Emergency assembly point opened.
60	Biomedical waste limit exceeded	BMW Coordinator, Transporter, SPCB	Auto-notify waste transporter for additional pickup. If daily generation significantly exceeds average → investigate unusual source department. SPCB reporting implications flagged.

6. Layer 4: Dynamic Form Builder

Every data entry screen in the system is built from configurable form templates. Hospital can modify any form without code changes.

Form Category	Configurable Forms (Examples — Each Hospital Customizes)
Registration Forms	New patient, emergency patient, newborn, MLC patient, VIP patient. Fields: configurable. ID proof types: configurable. Mandatory fields: configurable per patient type.
OPD Casesheets (per specialty)	General Medicine (16 systems review), Surgery (per-procedure), Pediatrics (growth chart, immunization), OB-GYN (ANC/PNC specific), Psychiatry (MSE format), Ophthalmology (visual acuity + slit lamp + fundus + IOP), ENT (ear diagram + nose + throat), Dermatology (body diagram lesion mapping), Orthopedics (joint examination + X-ray description), Dental (tooth chart), Community Medicine (family survey + epidemiological data). EACH specialty form has different fields, different layout, different scoring systems.
IPD Forms	Admission form, daily progress note (SOAP / narrative / problem-oriented), discharge summary, death summary, transfer note, referral letter, DAMA form.
Nursing Forms	Admission assessment (fall risk, Braden, nutritional), vital signs chart, I/O chart, medication administration record, nursing care plan, shift handover (SBAR), incident report, restraint monitoring, wound assessment (PUSH tool).
OT Forms	PAC assessment, WHO checklist (3 phases), operative note, anesthesia record, post-op order sheet, consent forms, specimen log, instrument count sheet.
Lab Forms	Sample collection form, rejection form, QC record, calibration record, EQAS result entry, critical value communication form.
Equipment Forms	PM checklist (per equipment type), calibration checklist, breakdown report, user training record, acceptance test checklist, condemnation form.
Administrative Forms	Leave application, purchase indent, work order, visitor pass, complaint form, employee onboarding checklist, appraisal form.

Form Builder Features: Drag-and-drop field placement. 20+ field types (text, number, date, dropdown, multi-select, radio, checkbox, image capture, body diagram annotation, signature, file upload, barcode scanner, calculated field, rich text, table/grid, rating scale, slider, toggle, GPS location, audio recording). Conditional logic (show/hide fields based on other field values). Validation rules (required, min/max, regex, custom formula). Section grouping with collapsible sections. Multi-page forms with progress indicator. Auto-save (prevent data loss). Offline capability (for mobile).

7. Layer 5: Rules Engine

The Rules Engine processes IF-THEN-ELSE business logic without code changes. Hospital administrators configure rules through a visual builder.

Category	Rule Examples	Parameters (Configurable)	Default Value
Clinical — Lab	Critical value alert threshold per test	Potassium critical range	< 2.5 OR > 6.5 mmol/L
Clinical — Lab	Auto-validation criteria	Tests eligible for auto-validation	List of low-risk tests
Clinical — Lab	Delta check (rate of change)	Hemoglobin delta threshold	> 3 g/dL change in 24hr = flag
Clinical — Pharmacy	Drug interaction severity override	Who can override critical interactions	Consultant only (not resident)
Clinical — Pharmacy	Antibiotic duration alert	Days before antibiotic review prompt	5 days (configurable per hospital antibiotic policy)
Clinical — Pharmacy	Renal dose adjustment trigger	Creatinine threshold for dose check	Creatinine > 1.5 mg/dL
Clinical — Nursing	Early warning score threshold	NEWS/MEWS escalation trigger	Score ≥ 5 = urgent review, ≥ 7 = critical outreach
Clinical — Nursing	Fall risk reassessment frequency	Hours between Morse scale reassessments	Every 12 hours (IPD), every shift change
Financial — Billing	Discount approval thresholds	Percentage tiers for approval chain	0-5% auto, 5-15% floor manager, 15-25% finance head, >25% CFO
Financial — Billing	Credit limit for cash patients	Maximum charges before payment required	INR 50,000 (configurable)
Financial — Purchase	Purchase approval thresholds	Amount tiers for approval authority	< 50K HOD, 50K-5L Purchase Committee, > 5L Management Board
Operational — Bed	Discharge prediction	Expected LOS per diagnosis for bed planning	Per DRG/ICD-10 category from historical data
Operational — Queue	Priority queue rules	Who gets priority in OPD queue	Emergency referral > Disabled > Elderly > Appointment > Walk-in
Equipment — PM	PM overdue escalation	Days overdue before escalation	7 days overdue = BME Head, 14 days = Medical Superintendent
Equipment — Calibration	Calibration grace period	Days after expiry before hard block	0 days for Class D (life-support), 7 days for Class C, 30 days for Class B
Compliance — NMC	Minimum OPD count alert	Daily OPD threshold per department	Medicine: 80, Surgery: 60 (NMC MSR norms)
Compliance — NABH	Quality indicator threshold	Target value per indicator	Hospital-acquired infection rate < 1%
Safety — Incident	Sentinel event auto-escalation	Time to Medical Superintendent notification	Within 1 hour of event documentation

8. Hospital Onboarding Framework

The 30-day onboarding process that takes a hospital from zero to go-live using pure configuration.

Day	Phase	Activities	Deliverables
1-2	Discovery	Hospital walkthrough, department inventory, current workflow documentation (shadow key staff), pain points, IT infrastructure assessment, building layout mapping.	Hospital profile document, current-state process map, infrastructure readiness report.
3-5	Topology Setup	Building/floor/wing/room/bed mapping in system. Network configuration. Printer mapping. Kiosk placement. MGPS outlet mapping. Fire equipment mapping.	Complete hospital topology configured. Bed master loaded. Zone map verified.
5-7	Department Setup	Department hierarchy created. Service catalog loaded (from hospital tariff book). Doctor profiles created. Department-ward-OT mapping. Formulary import.	Department tree live. Service catalog active. User accounts created for key staff.
7-10	Workflow Configuration	Default workflows reviewed with department heads. Customizations identified and configured. Approval chains set up. SLA values configured.	All clinical workflows configured. Key admin workflows configured.
10-14	Form Customization	Casesheet review per specialty with senior doctors. Custom fields added/removed. Nursing forms reviewed. Billing forms configured.	All clinical forms finalized. Specialty-specific casesheets approved by department heads.
14-18	Rules & Integration	Clinical rules configured (critical values, drug interactions, thresholds). Lab analyzer interfacing. Accounting system integration. SMS/WhatsApp setup. ABDM registration.	Rules engine active. Integrations tested. Communication channels verified.
18-22	Billing Setup	Tariff master loaded. Rate schemes created. Tax configuration. Insurance/TPA panel loaded. PMJAY/CGHS rates mapped. Package creation.	Billing system fully configured and verified.
22-25	Training	Role-based training: Receptionists (2 days), Doctors (1 day per department), Nurses (2 days), Lab (2 days), Pharmacy (1 day), Billing (2 days), Admin (1 day). Train-the-trainer for super-users.	Training completion records. Super-user identified per department.
25-28	Parallel Run	System runs alongside existing process. Data entered in both systems. Discrepancy identification and resolution. Performance optimization.	Parallel run report. Issue log. Performance benchmarks.
28-30	Go-Live	Cutover from old system. Legacy data migration (if applicable). Go-live support team on-site. Issue escalation hotline. Daily review meetings.	System live. Support handover. SLA monitoring active.

Post Go-Live Support: Month 1: On-site support team (2-3 people). Month 2-3: Remote support with weekly site visits. Month 4-6: Remote support. Month 7+: Standard AMC/SLA-based support.

Onboarding Team Requirement (From Our Side): 1 Project Manager, 1 Clinical Consultant (doctor/nurse with HMS experience), 1 Technical Consultant (integration, infrastructure), 1-2 Trainers. Total: 4-5 people per deployment for 30 days.

9. Document Summary & Cross-Reference

This RFC (RFC-HMS-2026-002) completes the specification trilogy:

Document	Focus	Key Contents
RFC-HMS-2026-001 v2.0	Core System + ABAC Security	67 modules overview, ABAC engine architecture, 8 access levels, 5 emergency tiers, 14 module-wise access matrices, department-wise equipment catalog, regulatory compliance, pricing.
RFC-HMS-2026-001 v2.1	Extended Modules	50 additional modules with full ABAC: Mortuary, Cath Lab, Endoscopy, Palliative, Nuclear Medicine, Psychiatry (MHCA), MGPS, BMW, Fire Safety, Water/Energy, Security, Visitor Mgmt, Feedback, Queue, Consent, Incident Reporting, Committees, Document Control, Communication, Anatomy, Teaching Labs, Simulation/OSCE, Research/IEC, Library, CME/FDP, Internship, Hostel, Student Welfare, Stipend/Bond, Alumni, NARF/NAAC.
RFC-HMS-2026-002 (This Document)	Workflow & Configuration Engine	7-layer configuration architecture, 120+ workflow templates (detailed step-by-step), 60+ cross-module automation links, dynamic form builder, rules engine, billing configurator, integration hub, 30-day hospital onboarding framework.

Together, these three documents define approximately: 67 modules, 120+ configurable workflows, 60+ cross-module automation links, 250+ ABAC access rules, 50+ role definitions, 500+ equipment items tracked, 800+ API endpoints, 200+ configurable forms, 100+ business rules — making this the most comprehensive HMS specification ever created for the Indian healthcare market.

Total system complexity is managed through 7 configuration layers, ensuring that the same codebase serves every hospital from a 20-bed rural center to a 2000-bed medical college — with zero code changes between deployments.